DSC 106

FINAL PROJECT PROPOSAL

PROJECT TITLE

Climate Change: Global Temperature Trends and Future Predictions

TEAM MEMBERS

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DATASET

CMIP6 (Coupled Model Intercomparison Project Phase 6) - CESM2-WACCM

RESEARCH QUESTION

How have temperature trends varied across world regions, and which regions are projected to experience the greatest warming in the future?

BRIEF WRITEUP

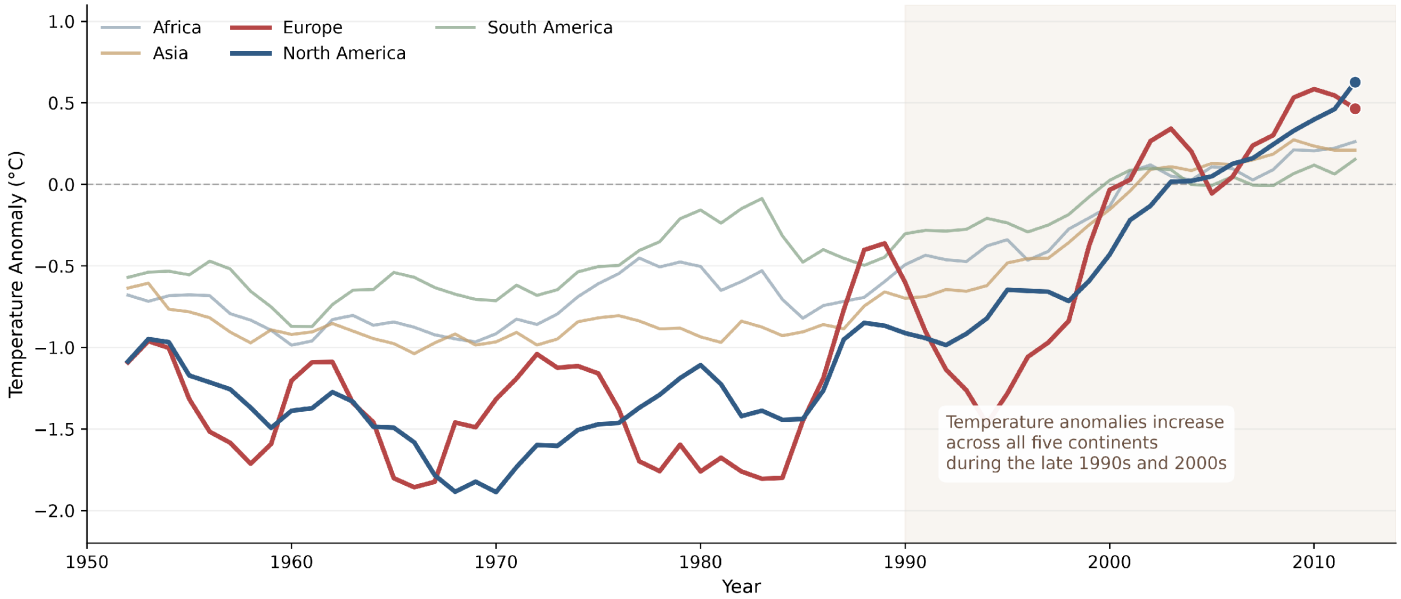
Our project investigates regional variations in historical and projected temperature changes using CMIP6 data from the CESM2-WACCM climate model. We analyze historical simulations (1950–2014) and future projections (2020–2099) under the SSP5-8.5 emissions scenario, focusing on geographic differences rather than comparisons across scenarios. Our analysis examines five continents: North America, South America, Europe, Africa, and Asia. Using line charts, bar charts, heatmaps, geographic maps, dumbbell charts, and scatter plots, we compare historical warming patterns, projected temperature increases, and changes in warming rates. We also investigate the relationship between latitude and projected warming, particularly at higher northern latitudes. Our final interactive website will allow users to select regions and explore temperature changes over time. Our goal is to communicate regional climate differences through spatial and temporal analysis. As our analysis uses a single climate model and emissions scenario for future projections, the results are model-specific rather than certain future outcomes.

PLANNED VISUALIZATIONS

01 Historical Temperature Trends
02 Historical Warming Ranking
03 Decadal Temperature Heatmap
04 Future Continental Warming Map
05 Projected Warming Rate Changes
06 Latitude vs. Projected Warming

Historical Temperature Anomalies Across Five Continents

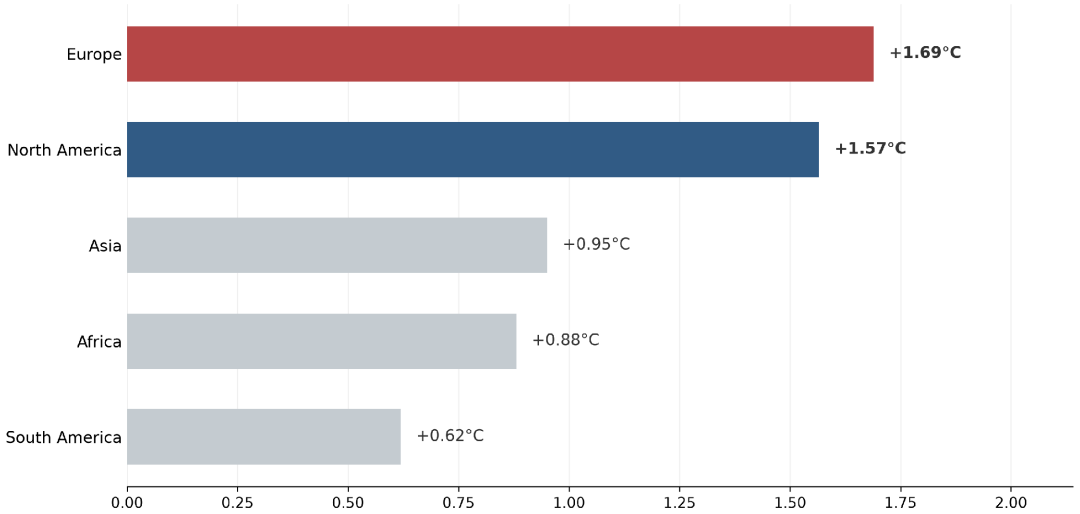
Five-year moving averages of annual temperature anomalies (1950–2014)



Source: CMIP6 · CESM2-WACCM historical simulation | Baseline: 1995–2014 | Land-area weighted continental means

Europe and North America Show the Largest Historical Warming

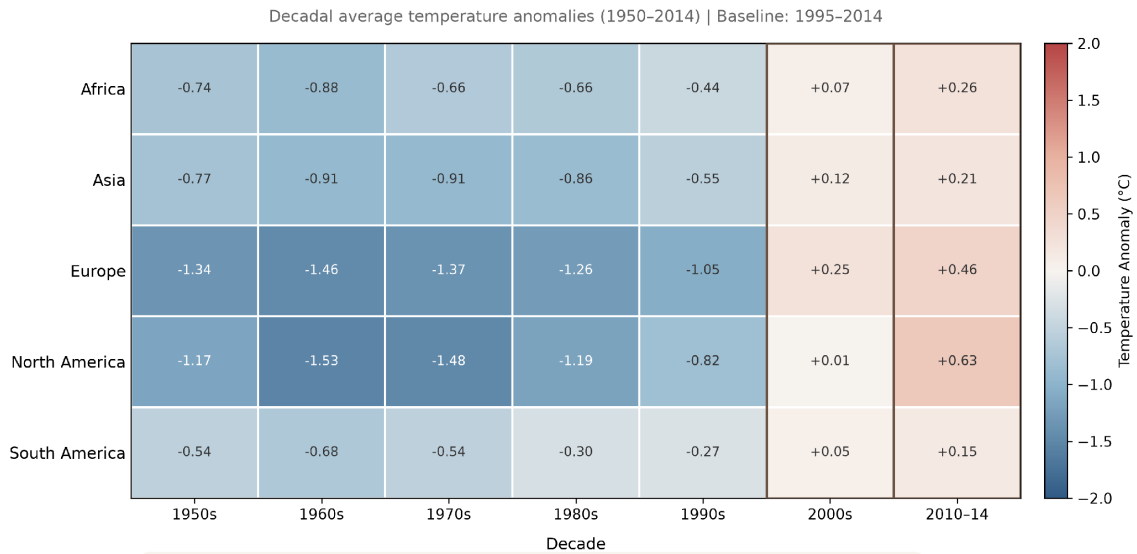
Change in average temperature: 1950–1959 vs. 2005–2014



KEY INSIGHT | Europe's warming was 2.7 times that of South America.

Source: CMIP6 · CESM2-WACCM historical simulation
Method: Land-area weighted continental means; five selected continents.

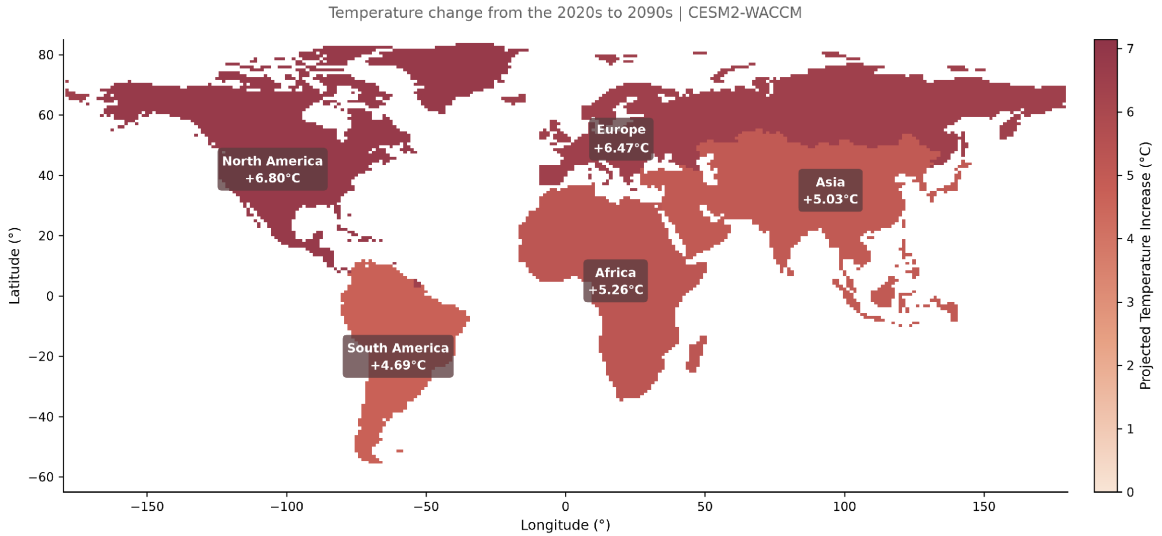
Continental Temperature Anomalies Shift Toward Warming by the 2000s



KEY INSIGHT | By the 2000s, all five selected continents had shifted to positive temperature anomalies.

Source: CMIP6 · CESM2-WACCM historical simulation
Method: Land-area weighted continental means | 2010–2014 is a partial decade.

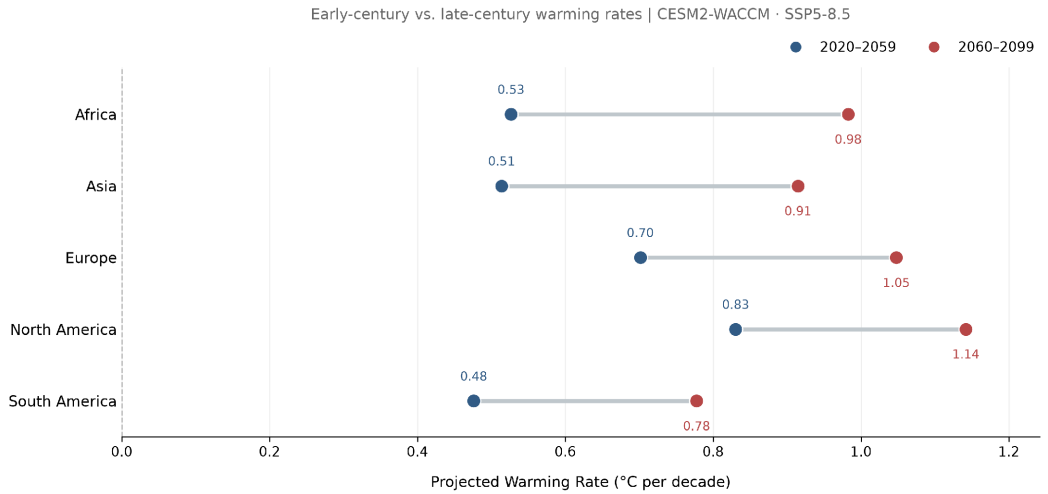
Projected Continental Warming Under SSP5-8.5



KEY INSIGHT | North America is projected to warm 2.11°C more than South America between the 2020s and 2090s.

Source: CMIP6 · CESM2-WACCM · SSP5-8.5
Method: Land-area weighted continental means | 2020–2029 vs. 2090–2099.
Note: Colors represent continent-level averages, not individual grid-cell projections.

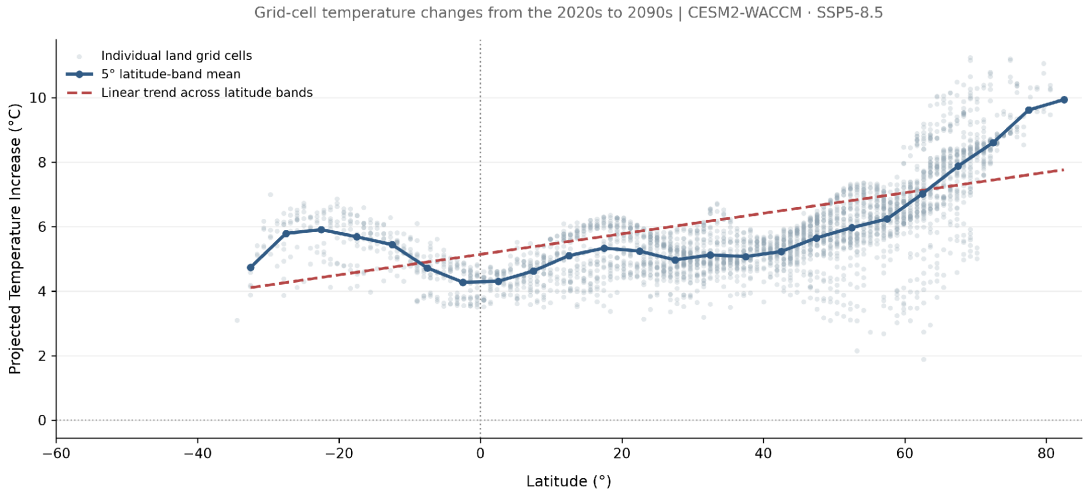
Projected Warming Rates Change Across Continents



KEY INSIGHT | Africa shows the largest increase in warming rate (+0.46°C/decade).

Source: CMIP6 · CESM2-WACCM · SSP5-8.5
Method: Land-area weighted continental means; linear trends estimated separately for 2020–2059 and 2060–2099.
Rates are expressed in °C per decade.

Higher Northern Latitudes Show Greater Projected Warming



KEY INSIGHT | Higher northern latitudes show greater projected warming (r = 0.72).

Source: CMIP6 · CESM2-WACCM · SSP5-8.5
Method: Grid-cell mean temperature change, 2020–2029 vs. 2090–2099; five selected continents.
Trend: Land-area-weighted 5° latitude-band means; descriptive linear fit.